Zero-Shot Industrial Anomaly Detection via Deep Feature Modeling

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1. Background and Motivation

Global manufacturing loses approximately $1.3 trillion annually to quality failures [1], with poor quality costing firms up to 20% of their total revenue [2]. In the high-speed era of Industry 4.0, traditional human inspection is no longer viable due to subjective bias, physical fatigue, and an inability to scale within automated environments. The shift toward Zero Defect Manufacturing requires a transition from reactive inspection to proactive, autonomous quality control.

A critical challenge in smart factories is that defects are rare, often representing less than 1% of total output. This extreme data imbalance makes it practically impossible to collect enough labeled "bad" images for standard supervised learning. Furthermore, the cost of an undetected anomaly is staggering; depending on the sector, faulty shipments or unplanned downtime can cost between $36,000 and $2 million per hour [3, 4]. Industry leaders now require systems capable of identifying abnormal states without having previously encountered a specific defect.

This project addresses these pain points through Zero-Shot Deep Feature Modeling using the MVTec AD dataset [5]. Instead of training a model to recognize specific defects like scratches or dents, we model the "distribution of perfection." By defining the statistical boundaries of a healthy product, our system can flag any deviation as an anomaly. This approach ensures robust quality control even when faced with novel defects, providing the high-fidelity precision necessary for modern industrial ecosystems.

2. Research Question

RQ1: How effectively can deep feature-based one-class models detect unseen industrial defects when trained only on normal samples, addressing the fundamental data scarcity problem in high-stakes manufacturing environments?

RQ2: How do probabilistic models (GMM), kernel methods (One-Class SVM), and tree-based methods (Isolation Forest) compare in balancing detection sensitivity and false-positive rates across the diverse categories of the MVTec AD dataset?

Hypothesis: We hypothesize that different methods will show distinct performance patterns based on data characteristics. GMM is expected to perform better in complex, multi-modal object categories due to its distributional flexibility, while One-Class SVM is expected to excel in simpler texture categories through strict boundary enforcement. Isolation Forest, with its tree-based ensemble structure, is hypothesized to achieve the highest average AUC-ROC across all categories by effectively isolating anomalies without relying on distributional assumptions, making it more robust to diverse defect types.

3. Dataset Description

The Project uses the MVTec AD dataset, a real-world benchmark designed for industrial anomaly detection for quality control in manufacturing industries. The dataset contains high-resolution RGB images of objects and textures related to industrial correspondence. The training sets consist only of defect-free or healthy samples, while only the test sets have anomalies along with defect-free samples. To ensure sufficient statistical modeling

capacity while maintaining computational feasibility, this study focuses on the three categories that have the largest number of training samples: carpet, hazelnut, and screw.

- **Sample Size:** The selected categories; carpet, hazelnut, and screw, contain training sets of the following counts: 280, 391, and 320 training images respectively. For instance, for the category "hazelnut" the test set has the crack, hole, cut, print categories of anomaly images, whereas the category "screw" has manipulated_front, scratch_head, scratch_neck, thread_side, thread_top. The total number of test set images are 117, 110, and 161 for carpet, hazelnut, and screw respectively.
- **Variables Used:** The input variables are high-dimensional deep feature embeddings of RGB image data, extracted using a pre-trained convolutional neural network, compressed via dimensionality reduction techniques.
- **Target Variable:** The primary target variable is a binary image-level anomaly label (normal vs. defective), with pixel-level ground truth masks available for defect segmentation performance assessment. **Target class is "Anomaly" (Defect).**

4. Methodology

Phase 1: Deep Feature Extraction and Dimensionality Reduction

In this first stage, unsupervised feature extraction is carried out using a convolutional neural network (CNN). We capture activations from the intermediate layers of the network to generate a high-dimensional feature vector for each defect-free training image. We then use an unsupervised dimensionality reduction technique, keeping a predetermined percentage of the variance, to lessen the curse of dimensionality and concentrate on the most notable features of the "normal" class. To build a condensed, information-rich representation of normal samples, the basis for ensuing one-class classification models is essential. Through initial testing, the precise CNN architecture and variance threshold will be established.

Phase 2: Anomaly Detection Models

We will implement and compare three distinct algorithms:

1. **Gaussian Mixture Models (GMM):** Model normal data as a mixture of Gaussians and use Mahalanobis distance to flag samples exceeding learned threshold as anomalies.
2. **One-Class SVM (OC-SVM):** Learn a kernel-based boundary around normal data using an RBF kernel, tuning γ (bandwidth) and ν (outlier fraction) to detect anomalies.
3. **Isolation Forest:** An ensemble tree method that isolates anomalies quickly through random feature splits; anomalies require fewer splits than normal points.

Phase 3: Evaluation

Since dataset consists of only good images, we will not use simple accuracy as part of evaluation. We will evaluate performance using Area under the Receiver Operating Characteristic Curve (AUC-ROC) and Precision-Recall curves to ensure our models are robust.

5. References

1. Siemens (2024). "The True Cost of Downtime." Fortune Global 500 Manufacturing Analysis.
2. American Society for Quality (2019). "Quality Costs: Definition and Estimation."
3. REWO (2025). "The True Cost of Downtime from Human Error in Manufacturing."
4. Senseye (2021). "The True Cost of Downtime Report."
5. Dataset - https://www.kaggle.com/datasets/ipythonx/mvtec-ad